

IEAP Series 03 — Data wrangling and ANOVA

RStudio assignment · Team SAUGE / LAGARDE / AHMAD

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- **GitHub repository:** <https://github.com/zoesauge/IEAP-Series03-RStudio>
- **Course:** IEAP, Series 03, *Data wrangling and ANOVA*
- **Article reproduced:** Mottet et al. (2017), *Trajectory formation principles are the same after mild or moderate stroke* [1].
- **How this file is built:** this is the *master document*. Each section lives in its own file in `sections/` and is pulled in with an `include`, so that each team member edits a different file.

1 Think before coding (1.1)

Scientific question. Is the effect of task difficulty (ID) on movement time the same in the three groups?

1. **How do we visualise it?** Mean MT ($\pm$ SD) as a function of ID, one line per group. If the effect of ID is the same in all groups, the lines are parallel.
2. **Is there a significant difference?** In a mixed ANOVA, the effect of ID differs between groups if the **GROUP $\times$ ID interaction** is significant. The main effects of GROUP and ID are secondary, and must be read with care if the interaction is significant [2].
3. **What sentence do we write?** A sentence that gives the direction of the effect and the statistics, $F(DFn, DFd)$, p and an effect size.

2 Which library? (1.2)

We use **ez** [3], the library suggested first in the assignment. Its three functions cover the whole analysis with the same arguments (data, dependent variable, participant identifier, between- and within-subjects factors):

- `ezANOVA()` runs the mixed ANOVA and returns the ANOVA table with the generalised eta squared (η^2) as effect size [4], [5] and, when needed, Mauchly's test of sphericity with the Greenhouse–Geisser and Huynh–Feldt corrections;
- `ezPlot()` plots the cell means, with error bars that show which means differ;
- `ezStats()` returns the descriptive statistics (N, mean, SD) of each cell.

Data reorganisation and figures use the **tidyverse** [6].

3 Data reorganisation and ANOVA (1.3)

3.1 Load the data

Each row of `Results.txt` is one movement: group, participant (SUBJ), trial, orientation (ORI), direction (DIR), index of difficulty (ID), repetition (REP), number of velocity peaks (`NbVelPeaks`, i.e. the number of sub-movements) and movement time in seconds.

```
library(tidyverse) # data wrangling and ggplot2
library(ez)        # mixed-design ANOVA (ezANOVA, ezPlot, ezStats)
library(here)      # paths relative to the project root
theme_set(theme_minimal(base_size = 11))
```

```
raw <- read.table(here("data", "Results.txt"), header = TRUE,
                  stringsAsFactors = TRUE)
glimpse(raw)
```

```
Rows: 7,560
Columns: 9
$ GROUP      <fct> ap, ap, ap, ap, ap, ap, ap, ap, ap, ap, ap, ap, ap, ap, a...
$ SUBJ       <fct> apS01, apS01, apS01, apS01, apS01, apS01, apS01, apS01, a...
$ TRIAL      <fct> T01, T01, T01, T01, T01, T01, T01, T01, T01, T01, T01, T0...
$ ORI        <fct> W, W, NW, NW, N, N, NE, NE, E, E, W, W, NW, NW, N, N, NE,...
$ DIR        <fct> outward, inward, outward, inward, outward, inward, outwar...
$ ID         <fct> ID1, ID1, ID1, ID1, ID1, ID1, ID1, ID1, ID1, ID1, ID1, ID2, ID...
$ REP        <fct> R1, R1, R1, R1, R1, R1, R1, R1, R1, R1, R1, R1, R1, R1, R...
$ NbVelPeaks <dbl> 2, 2, 3, 2, 3, 3, 1, 2, 4, 1, 2, 6, 3, 2, 4, 5, 6, 3, 3, ...
$ MovementTime <dbl> 1.23, 1.26, 1.14, 0.90, 1.11, 0.81, 1.08, 0.81, 1.32, 1.0...
```

3.2 Check for missing data

A mixed ANOVA needs every participant to have a value in every cell of the within-subjects factors. We count the movements in each combination of factors.

```
raw |> count(GROUP, ID)
```

	GROUP	ID	n
1	ap	ID1	740
2	ap	ID2	740
3	ap	ID3	740
4	cp	ID1	1140
5	cp	ID2	1140
6	cp	ID3	1140
7	pp	ID1	760
8	pp	ID2	760
9	pp	ID3	400

The group pp has only 400 movements for ID3, against 760 for ID1 and ID2: many patients did not perform ID3, the most difficult condition. We remove ID3.

```
d <- raw |> filter(ID != "ID3") |> droplevels()
levels(d$ID)
```

```
[1] "ID1" "ID2"
```

```
d |> count(GROUP, ID)
```

	GROUP	ID	n
1	ap	ID1	740
2	ap	ID2	740
3	cp	ID1	1140
4	cp	ID2	1140
5	pp	ID1	760
6	pp	ID2	760

We now check the repetitions.

```
d |> count(REP)
```

	REP	n
1	R1	2040
2	R2	2040
3	R3	1200

```
d |> count(GROUP, REP)
```

	GROUP	REP	n
1	ap	R1	520
2	ap	R2	520
3	ap	R3	440
4	cp	R1	760
5	cp	R2	760
6	cp	R3	760
7	pp	R1	760
8	pp	R2	760

The third repetition is missing for the group pp and incomplete for the group ap (440 instead of 520 movements). We remove R3.

```
d <- d |> filter(REP != "R3") |> droplevels()
d |> count(GROUP, REP)
```

	GROUP	REP	n
1	ap	R1	520
2	ap	R2	520
3	cp	R1	760
4	cp	R2	760
5	pp	R1	760
6	pp	R2	760

```
d |> distinct(GROUP, SUBJ) |> count(GROUP, name = "count")
```

	GROUP	count
1	ap	13
2	cp	19
3	pp	19

The final data set has 13 age-matched adults, 19 young adults and 19 patients (51 participants), which matches the 48 error degrees of freedom of the article ($51 - 3 \text{ groups} = 48$).

3.3 One value per participant and condition

The ANOVA must compare participants, not movements: otherwise the hundreds of movements of one participant would be treated as independent observations (pseudo-replication). We average the movements of each participant in each ID.

```
mt <- d |>
  group_by(GROUP, SUBJ, ID) |>
  summarise(MT = mean(MovementTime), .groups = "drop")
mt |> count(GROUP, ID)
```

```
# A tibble: 6 × 3
  GROUP ID      n
  <fct> <fct> <int>
1 ap    ID1     13
2 ap    ID2     13
3 cp    ID1     19
4 cp    ID2     19
5 pp    ID1     19
6 pp    ID2     19
```

3.4 Run the ANOVA

`ezANOVA()` takes the dependent variable (`dv`), the participant identifier (`wid`), the between-subjects factor and the within-subjects factor. We ask for type III sums of squares because the groups have unequal sizes (13, 19 and 19 participants).

```
aov_mt <- ezANOVA(data = mt, dv = MT, wid = SUBJ,
                  between = GROUP, within = ID, type = 3)
aov_mt$ANOVA
```

	Effect	DFn	DFd	F	p	p<.05	ges
2	GROUP	2	48	37.24408	1.720007e-10	*	0.57462110
3	ID	1	48	26.10886	5.543011e-06	*	0.06581307
4	GROUP:ID	2	48	11.58124	7.866206e-05	*	0.05882293

```
aov_tab <- aov_mt$ANOVA
```

With only two levels, the within-subjects factor ID cannot violate sphericity, so `ez` does not report Mauchly's test and no Greenhouse–Geisser correction is needed.

3.5 Visualise the interaction

`ezPlot()` draws the cell means. Its error bars are Fisher's Least Significant Difference (FLSD) bars: two means whose bars do not overlap differ significantly.

```
ezPlot(data = mt, dv = MT, wid = SUBJ, between = GROUP, within = ID,
       x = ID, split = GROUP, type = 3,
       x_lab = "ID", y_lab = "Movement Time (s)")
```

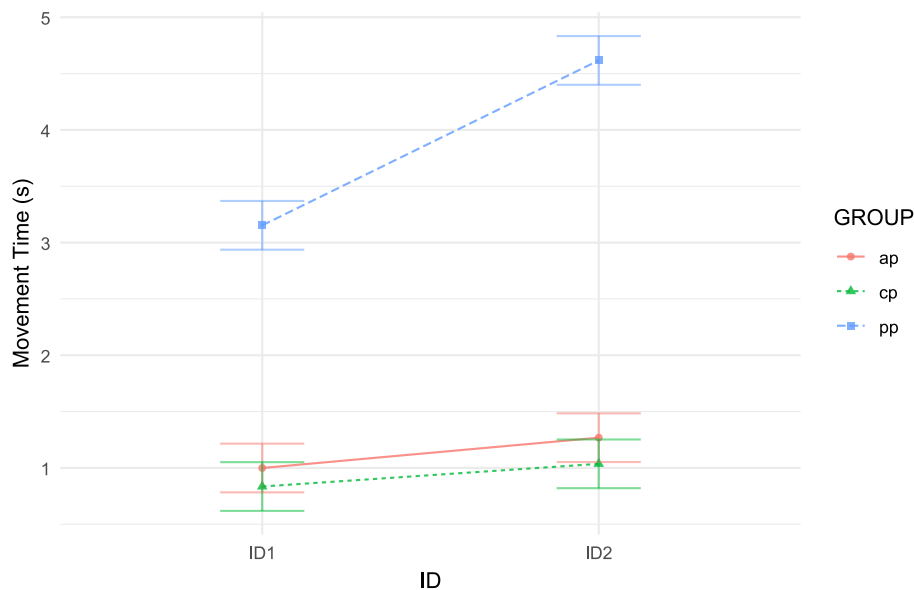


Figure 1: Movement time as a function of ID in each group (ezPlot; error bars: Fisher's Least Significant Difference). The lines are clearly not parallel: the effect of ID is much larger for the patients (pp).

ezStats() gives the descriptive statistics of each cell (number of participants, mean, SD and FLSD). We keep them for the next questions.

```
mt_summary <- ezStats(data = mt, dv = MT, wid = SUBJ,
                      between = GROUP, within = ID, type = 3) |>
  rename(n = N, mean = Mean, sd = SD) |>
  as_tibble()
mt_summary |> mutate(across(c(mean, sd, FLSD), \(x) round(x, 2)))
```

```
# A tibble: 6 × 6
  GROUP ID      n mean  sd  FLSD
<fct> <fct> <dbl> <dbl> <dbl> <dbl>
1 ap    ID1      13  1    0.2  0.43
2 ap    ID2      13 1.27 0.29 0.43
3 cp    ID1      19 0.83 0.12 0.43
4 cp    ID2      19 1.04 0.11 0.43
5 pp    ID1      19 3.15 1.47 0.43
6 pp    ID2      19 4.62 2.41 0.43
```

3.6 Are the assumptions met?

ezANOVA() reports Levene's test only for purely between-subjects designs, so we run it at each ID with car::leveneTest() (installed with ez). We check normality in each cell with the Shapiro-Wilk test.

```
mt |> # equal variances between groups
  group_by(ID) |>
  group_modify(\(df, key) {
    lt <- car::leveneTest(MT ~ GROUP, data = df)
    tibble(F = lt$`F value`[1], DFn = lt$Df[1], DFd = lt$Df[2], p = lt$`Pr(>F)`[1])
  })
```

```
# A tibble: 2 × 5
# Groups:   ID [2]
```

	ID	F	DFn	DFd	p
	<fct>	<dbl>	<int>	<int>	<dbl>
1	ID1	21.8	2	48	0.000000184
2	ID2	12.2	2	48	0.0000524

```
mt |> # normality in each cell
  group_by(GROUP, ID) |>
  summarise(W = shapiro.test(MT)$statistic, p = shapiro.test(MT)$p.value,
            .groups = "drop")
```

```
# A tibble: 6 × 4
  GROUP ID      W      p
  <fct> <fct> <dbl> <dbl>
1 ap    ID1    0.926 0.306
2 ap    ID2    0.785 0.00457
3 cp    ID1    0.960 0.580
4 cp    ID2    0.922 0.122
5 pp    ID1    0.945 0.321
6 pp    ID2    0.881 0.0223
```

The variances are not equal between groups (Levene's test, $p < .001$ at both IDs; the SD of the patients is more than ten times that of the young adults), and movement time departs from normality in two of the six cells (ID2 for the age-matched adults and for the patients, Shapiro–Wilk $p < .05$). With unequal group sizes, this makes the p-values approximate. However, the effects are so large ($p < .001$ for all three effects) that this does not change the conclusion; a log transformation of MT, which reduces the skewness, would be a reasonable robustness check.

3.7 Result sentence

Movement time increased with the index of difficulty ($F(1, 48) = 26.11$, $p < .001$, $\eta_G^2 = 0.07$) and was longer for the patients than for both control groups ($F(2, 48) = 37.24$, $p < .001$, $\eta_G^2 = 0.57$). Crucially, the effect of difficulty was not the same in all groups (GROUP $\times$ ID interaction, $F(2, 48) = 11.58$, $p < .001$, $\eta_G^2 = 0.06$): from ID1 to ID2, movement time increased by 1.47 s in the patients (3.15 to 4.62 s), but only by 0.27 s in the age-matched adults and 0.21 s in the young adults.

4 Run the linear regression by GROUP

To investigate the relationship between movement time and task difficulty, we fit one linear regression for each group. The dependent variable is movement time (MT) and the predictor is the index of difficulty (ID).

The regression is performed on the participant-level means created in the previous section. This avoids treating the repeated movements of one participant as independent observations

4.1 Prepare the data

ID is a categorical variable with values ID1 and ID2. We create a numeric version for the regression, where ID1 becomes 1 and ID2 becomes 2.

```
#| label: regression-data

mt_regression <- mt |>
  mutate(
    ID_numeric = as.numeric(sub("ID", "", ID))
  )
```

```
mt_regression
```

```
# A tibble: 102 × 5
  GROUP SUBJ ID      MT ID_numeric
  <fct> <fct> <fct> <dbl>      <dbl>
1 ap    apS01 ID1    1.01         1
2 ap    apS01 ID2    1.30         2
3 ap    apS02 ID1    1.05         1
4 ap    apS02 ID2    1.34         2
5 ap    apS03 ID1    0.835        1
6 ap    apS03 ID2    1.17         2
7 ap    apS04 ID1    1.21         1
8 ap    apS04 ID2    1.58         2
9 ap    apS05 ID1    1.49         1
10 ap   apS05 ID2    2.10         2
# i 92 more rows
```

4.2 Fit one regression per group

```
regression_models <- mt_regression |>
  group_by(GROUP) |>
  group_modify(\(df, key) {
    model <- lm(MT ~ ID_numeric, data = df)

    tibble(
      intercept = unname(coef(model)[1]),
      slope = unname(coef(model)[2]),
      r_squared = summary(model)$r.squared,
      p_value = summary(model)$coefficients[2, 4]
    )
  }) |>
  ungroup()

regression_models
```

```
# A tibble: 3 × 5
  GROUP intercept slope r_squared    p_value
  <fct>      <dbl> <dbl>      <dbl>      <dbl>
1 ap         0.730 0.269      0.235 0.0120
2 cp         0.634 0.201      0.447 0.00000451
3 pp         1.69  1.46       0.124 0.0301
```

The regression equation is:

[$MT = a + b \text{ ID}$]

where (a) is the intercept and (b) is the slope. The slope indicates the increase in movement time associated with a one-unit increase in task difficulty. Therefore, the group with the largest slope is the group most affected by increasing task difficulty.

```
equations <- regression_models |>
  mutate(
    equation = sprintf(
```

```

      "MT = %.3f %s %.3f x ID",
      intercept,
      if_else(slope >= 0, "+", "-"),
      abs(slope)
    )
  )

equations |>
  select(GROUP, equation, r_squared, p_value)

```

```

# A tibble: 3 × 4
  GROUP equation          r_squared p_value
<fct> <chr>          <dbl>    <dbl>
1 ap    MT = 0.730 + 0.269 x ID    0.235 0.0120
2 cp    MT = 0.634 + 0.201 x ID    0.447 0.00000451
3 pp    MT = 1.689 + 1.464 x ID    0.124 0.0301

```

The regression slopes can be compared with the ANOVA interaction. A significant GROUP × ID interaction means that the effect of difficulty differs between groups. The group with the steepest regression line should show the largest increase in movement time from ID1 to ID2.

5 Save a graph in a PDF file for publication

The following graph shows the participant-level mean movement times, the regression line for each group, and the equation of each regression.

```

publication_plot <- ggplot(
  mt_regression,
  aes(
    x = ID_numeric,
    y = MT,
    colour = GROUP,
    group = GROUP
  )
) +
  geom_point(
    alpha = 0.25,
    size = 1.8
  ) +
  geom_smooth(
    method = "lm",
    formula = y ~ x,
    se = FALSE,
    linewidth = 0.9
  ) +
  geom_text(
    data = equation_labels,
    aes(
      x = x,
      y = y,
      label = equation,
      colour = GROUP
    ),
    inherit.aes = FALSE,
    hjust = 0,

```



```

    show.legend = FALSE
  ) +
  geom_blank(
    data = tibble(
      ID_numeric = c(0, 3),
      MT = c(0, 6)
    ),
    aes(x = ID_numeric, y = MT),
    inherit.aes = FALSE
  ) +
  scale_x_continuous(
    breaks = c(1, 2),
    labels = c("ID1", "ID2"),
    limits = c(0, 3)
  ) +
  scale_y_continuous(
    limits = c(0, 6)
  ) +
  labs(
    title = "Movement time as a function of task difficulty",
    x = "Index of difficulty (ID)",
    y = "Mean movement time (s)",
    colour = "Group"
  ) +
  theme_classic() +
  theme(
    plot.title = element_text(hjust = 0.5),
    legend.position = "right"
  )
publication_plot

```

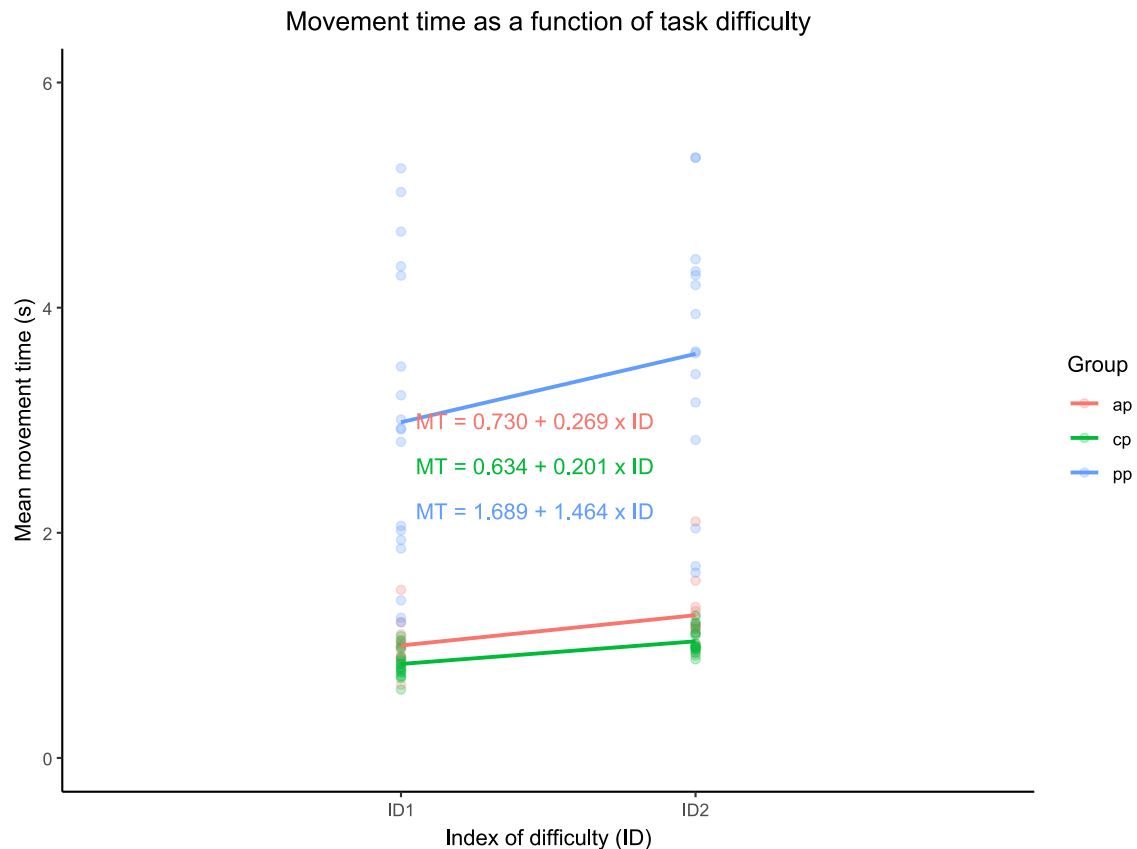


Figure 2: Mean movement time as a function of index of difficulty, with separate linear regression lines and equations for the three groups.

The lines are not parallel: the increase in movement time between ID1 and ID2 is larger for some groups (the stroke survivors) than for others (the healthy age-matched and young participants). This graphical result illustrates the significant $\text{GROUP} \times \text{ID}$ interaction found in the ANOVA.

6 Export the graph as a PDF file for publication

The graph is exported with a width of 8 inches and a height of 6 inches, as requested in the assignment.

```
ggsave(
  filename = "movement_time_regression.pdf",
  plot = publication_plot,
  width = 8,
  height = 6,
  units = "in",
  device = cairo_pdf
)
```

7 Differences with the article (1.7)

7.1 Putting our equations on the same scale as the article

In the regression section, ID was coded 1 for ID1 and 2 for ID2 (ID_numeric). The slopes of regression_models are therefore in seconds **per ID step**, and the intercepts are the movement times predicted at a step “0” that does not exist. The article fits Fitts’ law on the index of difficulty in **bits** [1], [7]. The two IDs of the task are 3.80 and 5.43 bits (x-axis of Fig 3 of the article), so one ID step is 1.63 bits.

To compare our equations with those of the article, we convert them to bits. The line goes through the same two group means; only the unit of the x-axis changes:

- slope (s/bit) = slope (s/step) / 1.63;
- intercept (s) = predicted MT at ID1 – slope (s/bit) × 3.80.

```
id_bits <- c(ID1 = 3.80, ID2 = 5.43)
step_bits <- unname(id_bits["ID2"] - id_bits["ID1"])

article <- tibble(
  GROUP = c("pp", "ap", "cp"),
  intercept_article = c(0.422, 0.622, 0.475),
  slope_article     = c(0.672, 0.118, 0.097)
)

comparison <- regression_models |>
  mutate(GROUP = as.character(GROUP),
         slope_ours = slope / step_bits,
         intercept_ours = (intercept + slope * 1) - slope_ours * id_bits["ID1"]) |>
  select(GROUP, intercept_ours, slope_ours, r_squared) |>
  left_join(article, by = "GROUP") |>
  mutate(GROUP = factor(GROUP, levels = c("pp", "ap", "cp"))) |>
  arrange(GROUP) |>
  mutate(slope_ratio_ours = slope_ours / slope_ours[GROUP == "cp"],
         slope_ratio_article = slope_article / slope_article[GROUP == "cp"])

comparison |> mutate(across(where(is.numeric), \(x) round(x, 3)))
```

```
# A tibble: 3 × 8
  GROUP intercept_ours slope_ours r_squared intercept_article slope_article
<fct>      <dbl>      <dbl>    <dbl>         <dbl>         <dbl>
1 pp        -0.259      0.898    0.124         0.422         0.672
2 ap         0.372      0.165    0.235         0.622         0.118
3 cp         0.366      0.123    0.447         0.475         0.097
# i 2 more variables: slope_ratio_ours <dbl>, slope_ratio_article <dbl>
```

7.2 What differs

- **The patients' slope is steeper:** 0.898 s/bit against 0.672 s/bit in the article, and the intercept becomes negative (-0.259 s), which has no physical meaning: a movement cannot take a negative time at ID = 0.
- **The controls' slopes are a little steeper too:** 0.165 and 0.123 s/bit, against 0.118 and 0.097 s/bit.
- **Our patients are slower at ID2:** their mean movement time is 4.62 s, with a very large spread between patients. This is why the patients' regression explains little of the variance ($R^2 = 0.12$): the difference between patients is much larger than the effect of ID.

These differences come from the simpler method used here. Several choices may matter (the article does not describe all of them, so these are hypotheses):

1. **Data selected.** We removed ID3 and the third repetition to obtain a complete design. The article may have kept them, or handled the missing cells differently.
2. **How trials are averaged.** We used the mean of all the movements of each participant. Movement time is right-skewed (a few very slow corrective movements), so the mean is pulled up by the slowest movements, especially in the patients. A median, a log transformation or the removal of outlier trials would give smaller values and a smaller slope.

3. **Which ID.** We used the *nominal* ID of the task. Fitts' law is often fitted on the *effective* ID, computed from the actual spread of the endpoints, which can differ between groups.
4. **The full design.** The article also includes orientation and direction as factors (with Greenhouse–Geisser corrections), whereas we averaged over them.
5. **The regression model.** Our regression treats the two values of each participant as independent observations, while the article analyses them as repeated measures. This does not change the slope (the line goes through the two group means), but it makes the R^2 and the p-values of the regressions approximate.

7.3 Are the conclusions the same?

```
# Do the two control groups differ? Same ANOVA without the patients.
mt_controls <- mt |> filter(GROUP != "pp") |> droplevels()
aov_controls <- ezANOVA(data = mt_controls, dv = MT, wid = SUBJ,
                        between = GROUP, within = ID, type = 3)$ANOVA
aov_controls
```

	Effect	DFn	DFd	F	p	p<.05	ges
2	GROUP	1	30	9.881641	3.744417e-03	*	0.232175364
3	ID	1	30	155.729390	2.084714e-13	*	0.298551219
4	GROUP:ID	1	30	3.236146	8.209810e-02		0.008767102

Yes, the conclusions are the same, because they rest on the *pattern* of the results, which is identical, and not on the exact values:

- **Movement time increases with difficulty in all groups:** the three slopes are positive and the main effect of ID is significant.
- **Patients are much slower** (main effect of GROUP, the largest effect, $\eta_G^2 = 0.57$).
- **Patients are much more sensitive to accuracy constraints:** their slope is 7.3 times that of the young adults here, and 6.9 times in the article. The ratio does not depend on how ID is coded, so it can be compared directly. The GROUP $\times$ ID interaction is significant ($p < .001$), so it passes even the stricter $\alpha = .01$ used in the article.
- **Age alone does not change the effect of difficulty:** the slopes of the two control groups are close in both analyses. Without the patients, the GROUP $\times$ ID interaction is not significant at $\alpha = .01$ ($p = 0.082$). The age-matched adults are slightly slower overall than the young adults (main effect of GROUP, $p = 0.004$), but this difference of about 0.2 s is tiny compared with the 2.8 s that separate the patients from the age-matched adults: the large difference is due to the stroke, not to age.

The effects are very large compared with the noise, so different reasonable choices in the data processing change the size of the estimates but not their direction or their significance. The main limit of our simpler analysis is that, with two IDs only, it cannot show that the relation between movement time and ID is *linear*; it only shows that movement time increases with ID, and much more steeply after a stroke.

Our Git workflow strategy

Repository. Zoé created the public repository IEAP-Series03-RStudio on her GitHub account and added Alexis and Albatoul as collaborators. Her first two commits set up the project (README, LICENSE, .gitignore, RStudio project, master document, references and data). After that, nobody committed directly on main: it only received merges.

One file per person. The master document only contains the header and the include lines. Each question lives in its own file in sections/, and each file has one owner, so we never edit the same lines and we never had a merge conflict.

Member	Questions	File	Branch
Zoé SAUGE	1.1, 1.2	_01-question.qmd	1.1Think
Alexis LAGARDE	1.3	_02-data-anova.qmd	1.2-data-anova
Albatoul AHMAD	1.4, 1.5, 1.6	_03-regression-graph.qmd	1.4-1.6-regression-graph
Zoé SAUGE	1.7	_04-discussion.qmd	1.7-discussion
Alexis LAGARDE	Git workflow	_05-git-workflow.qmd	git-challenges
Everyone	challenges, sources, checklist	_06-challenges-checklist.qmd	git-challenges

Branches and pull requests. Each part was developed on its own branch, created from an up-to-date main, and merged back through a pull request on GitHub.

Order of the merges. The questions depend on each other: the regression (1.4) uses the participant means `mt` created in 1.3, the figure (1.5–1.6) uses the regression, and the discussion (1.7) uses `regression_models`, `mt_summary` and the ANOVA table. The pull requests were therefore merged in the order of the questions: 1.1Think (#1), then 1.2-data-anova (#2), then 1.4-1.6-regression-graph, then 1.7-discussion. Before starting a new branch, each member pulled main.

Commit messages. Each commit does one step of a question and its message describes it in the imperative: for example Remove incomplete patient condition from ANOVA, Add per-group regression analysis section, Add PDF export for regression graph. Each question has several commits (4 for 1.1–1.2, 13 for 1.3, 7 for 1.4–1.6).

What we would do differently. Our branch names are not fully consistent: 1.1Think has no separator, and 1.2-data-anova contains question 1.3. A fixed pattern decided at the start (for example <question>-<topic>) would make the history easier to read.

Challenges encountered and lessons learned

Technical challenges

- **Incomplete data.** The data set looked complete at first sight (7,560 movements), but counting the movements per combination of factors showed that ID3 and the third repetition were missing for some groups. Lesson: always count the observations per cell before running an ANOVA, because a mixed ANOVA needs every participant in every cell.
- **Averaging before the ANOVA.** Running the ANOVA on single movements would treat each movement as an independent participant and give absurdly small p-values. We had to aggregate to one value per participant and condition first (`mt`).
- **A numeric ID.** ID is a factor in the ANOVA but must be a number for the regression. We coded it 1 and 2 (`ID_numeric`), which gives slopes per ID step. To compare our equations with the article, which uses ID in bits, we had to convert them in the discussion (1.7). Lesson: check the units of a model before comparing it with a published one.
- **The publication figure.** Combining the participant points, one regression line per group and the equations on the same graph required one data frame per layer (`mt_regression` for the points and lines, `equation_labels` for the text) and fixed axis limits.
- **Included files.** A section file that keeps its own YAML header (`title`, `format`) overrides the header of the master document, and the whole report was then rendered in HTML under the wrong title. Section files must contain only text and code.

Individual lessons

- **Zoé SAUGE:** I learned to start from the scientific question and break it into precise sub-questions before writing any code, and to choose a suitable library, such as `ez`, to run the ANOVA reliably. In the discussion, comparing our results with the article taught me to put results on the same scale before comparing them (ID steps vs. bits), and that a simpler method can lead to the same conclusions as long as its limits are clearly stated.
- **Alexis LAGARDE:** Better handling of the method to work with a section folder with a master document where we include all files in it. Working on 1.3 taught me to check the data before testing: counting the cells revealed missing conditions, and averaging per participant avoided pseudo-replication. On the Git side, small commits on one branch per question made teamwork easy, but we should have agreed on branch names and reviewed each other's pull requests.
- **Albatoul AHMAD:** I learned how to read research papers and actually understand what the figures are showing and how the researchers reached their results. I also learned how to use ANOVA to analyze the data and see whether there are meaningful differences between groups.

Data sources

Data	Description	Source
data/Results.txt	7,560 pointing movements: 51 participants (19 patients after stroke, 13 age-matched adults, 19 young adults), 3 IDs, 3 repetitions, 5 orientations, 2 directions; number of velocity peaks and movement time	Course Moodle page, IEAP Series 03: https://moodle.umontpellier.fr/pluginfile.php/3002188/mod_folder/content/0/Results.txt?forcedownload=1
Fig 3 of the article	ID values (3.80 and 5.43 bits) and regression equations used in 1.7	Mottet et al. (2017), PLoS ONE: https://doi.org/10.1371/journal.pone.0173674

Software: R, with the packages `tidyverse` [6], `ez` [3] and `here`.

Grading checklist

We went through this list before the final merge into `main`.

	Criterion	Where / how we checked
☐	Public repository named exactly IEAP-Series03-RStudio	https://github.com/zoesaugue/IEAP-Series03-RStudio , opened in a private browser window
☐	main branch up to date (it is the graded branch)	All pull requests merged; nothing left on other branches
☐	Clear and complete README.md	Team, structure, how to render, Git workflow
☐	LICENSE file	MIT license
☐	Master document IEAP-Series03-RStudio.qmd	Renders to PDF without error from a fresh clone
☐	Corresponding PDF IEAP-Series03-RStudio.pdf in the repository	Committed, at the end

	Criterion	Where / how we checked
☐	Authors and date in the header	Title page
☐	Clickable link to the GitHub repository	"About this report", first page
☐	Analysis of our Git workflow strategy	Section "Our Git workflow strategy"
☐	Challenges encountered and lessons learned	Section "Challenges", with one paragraph per member
☐	Complete code and answers to questions 1.1 to 1.7	One section per question, code visible
☐	Figure exported as an 8 × 6 inch PDF	movement_time_regression.pdf
☐	Data sources with URLs	Section "Data sources"
☐	Scientific references with DOIs	Bibliography
☐	Meaningful commit messages, at least one commit per question	Github commit history
☐	Principled use of branches	One branch per part, merged by pull request with a merge commit
☐	Each member committed from their own account	GitHub commit history
☐	PDF uploaded on Moodle	Done by: Zoé Sauge

Bibliography

- [1] D. Mottet, L. E. H. van Dokkum, J. Froger, A. Gouaich, and I. Laffont, "Trajectory formation principles are the same after mild or moderate stroke," *PLoS ONE*, vol. 12, no. 3, p. e173674, 2017, doi: 10.1371/journal.pone.0173674.
- [2] A. Field, J. Miles, and Z. Field, *Discovering Statistics Using R*. London: SAGE Publications, 2012.
- [3] M. A. Lawrence, "ez: Easy Analysis and Visualization of Factorial Experiments." 2026. doi: 10.32614/CRAN.package.ez.
- [4] S. Olejnik and J. Algina, "Generalized eta and omega squared statistics: Measures of effect size for some common research designs," *Psychological Methods*, vol. 8, no. 4, pp. 434–447, 2003, doi: 10.1037/1082-989X.8.4.434.
- [5] R. Bakeman, "Recommended effect size statistics for repeated measures designs," *Behavior Research Methods*, vol. 37, no. 3, pp. 379–384, 2005, doi: 10.3758/BF03192707.
- [6] H. Wickham *et al.*, "Welcome to the tidyverse," *Journal of Open Source Software*, vol. 4, no. 43, p. 1686, 2019, doi: 10.21105/joss.01686.
- [7] P. M. Fitts, "The information capacity of the human motor system in controlling the amplitude of movement," *Journal of Experimental Psychology*, vol. 47, no. 6, pp. 381–391, 1954, doi: 10.1037/h0055392.